

Introduction to Database

Simple Introduction

What is a Database?

- A database is an organized collection of data, generally stored and accessed electronically from a computer system.

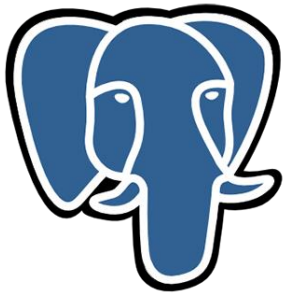
Refer https://en.wikipedia.org/wiki/Database

Some DBMS Products









PostgreSQL





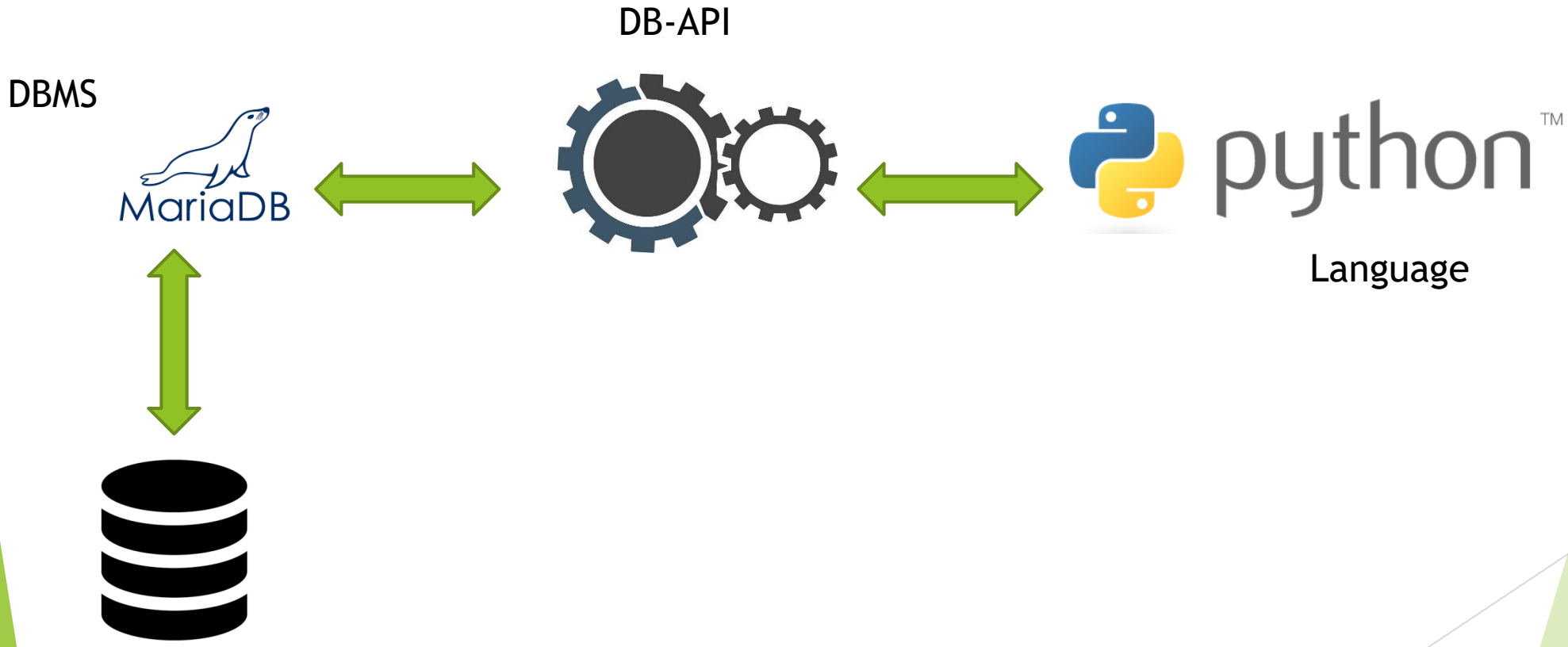


Interbase

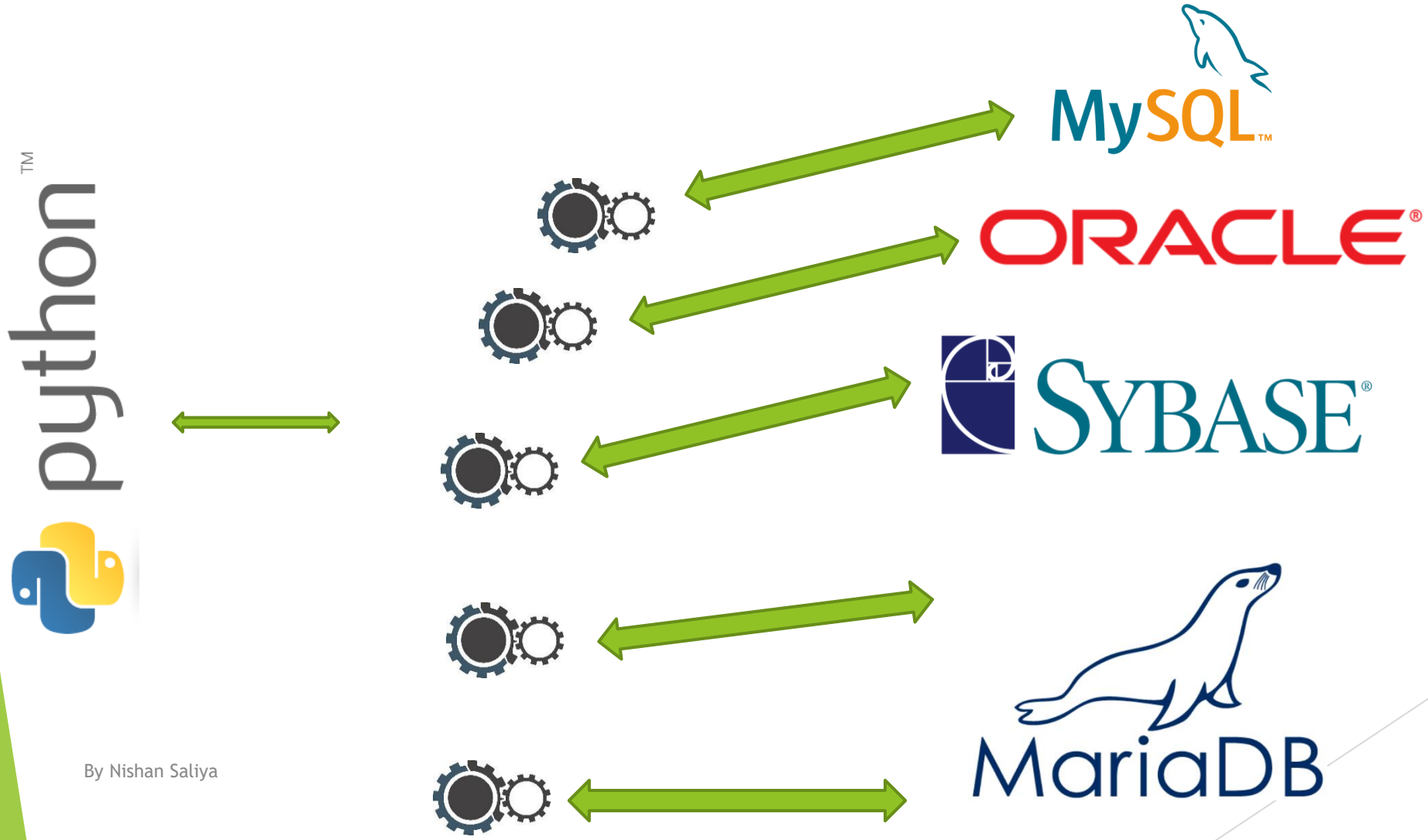
What is DB-API?

- DB-API means “Database Application Programming Interface”
- This will work as a connector between the Python and the Database product

How DB-API Works?



Different DB-APIs for Various Products



Our Product

- We will be using the below products for our database activities







What You Need?

- Python programming language
  - https://www.python.org/downloads/
- MySQL database system
  - https://www.apachefriends.org/download.html
- MariaDB Connector

Install MariaDB Connector

- Open the command console in Administrator mode (better with this mode)
- Make sure you are connected to the internet
- Type the below command and press the enter key.

```
pip install mariadb
```

- Observe the installation

Other Versions for this Command

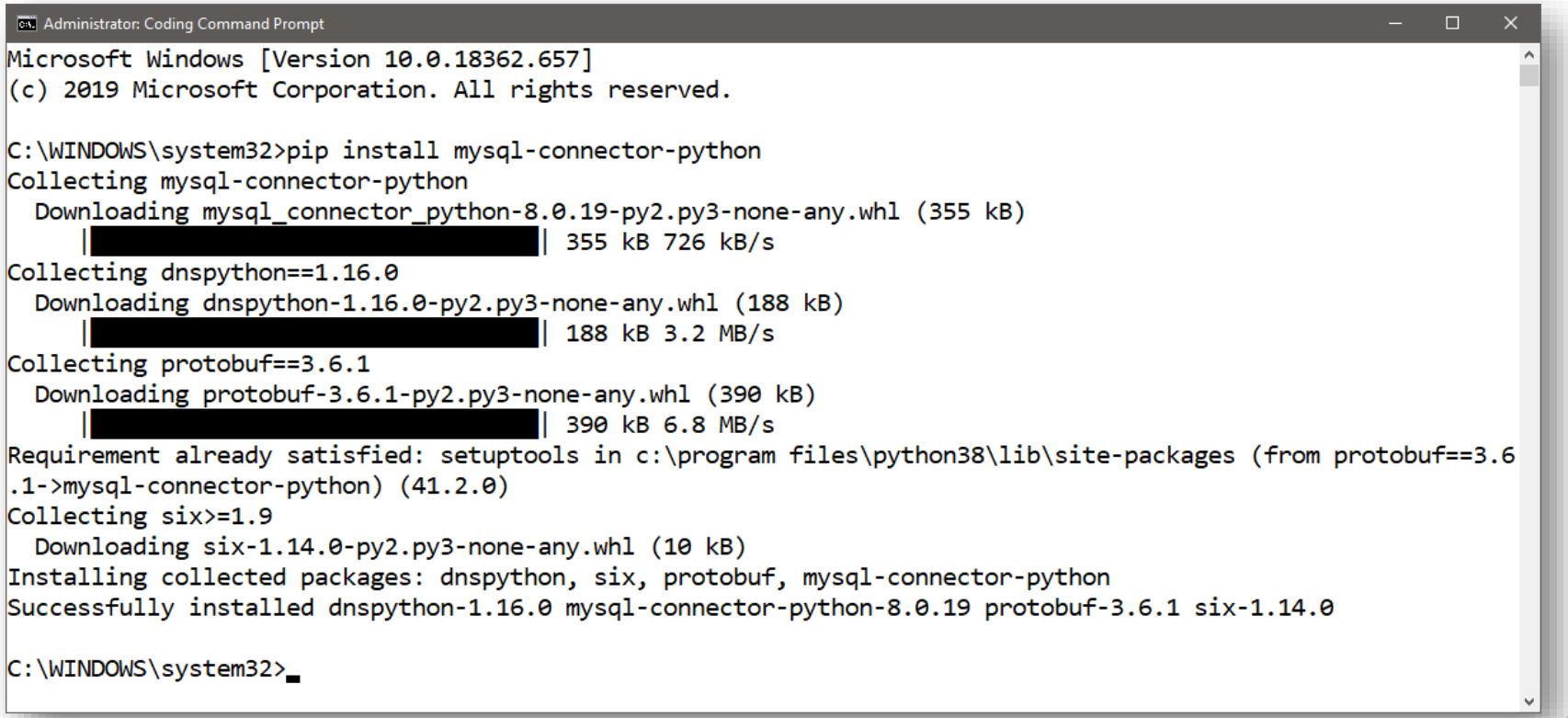
PCs

- pip install mariadb
- py -m pip install mariadb
- python -m pip install mariadb

Mac Machines

- pip3 install mariadb
- py -m pip3 install mariadb
- python -m pip3 install mariadb

MariaDB Connector



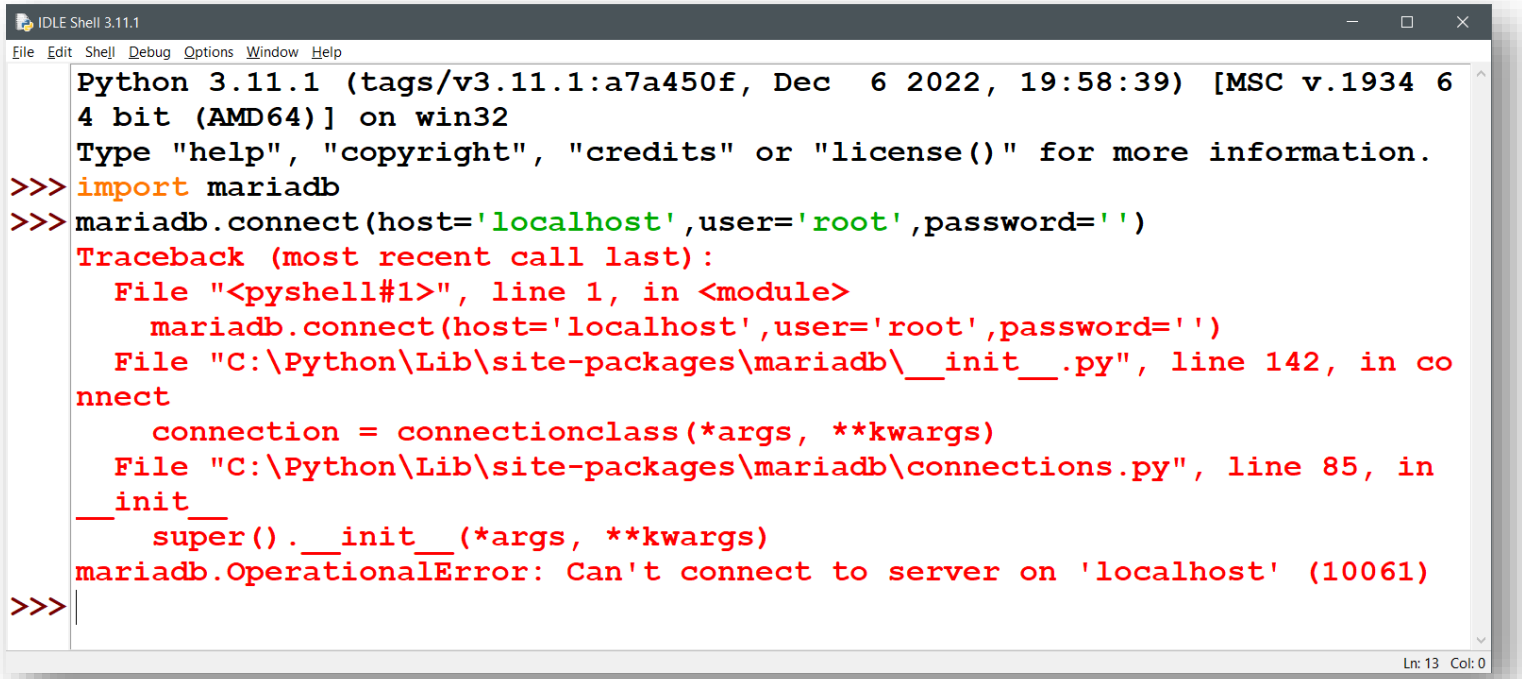
Test Your Python + MariaDB Connection

- Open IDLE and try the below two command codes

```
>>> import mariadb
>>> mariadb.connect(host='localhost',user='root',password='')
<mariadb.connection connected to 'localhost' at 000001E12E011010>
>>>
```

- If all is working, you must be able to see blue color output line as above

Common Issues with Connection



Reasons may be

1. Database Server is not running
2. Wrong computer name
3. Wrong username
4. Wrong password
5. Other …